

Birthday Problem

Green Book 2.2: Logic Reasoning

Problem. You and your colleagues know that your boss A 's birthday is one of the following 10 dates:

Mar 4, Mar 5, Mar 8
Jun 4, Jun 7
Sep 1, Sep 5
Dec 1, Dec 2, Dec 8

A told you only the month of his birthday, and told your colleague C only the day. After that, you first said: "I don't know A 's birthday; C doesn't know it either." After hearing what you said, C replied: "I didn't know A 's birthday, but now I know it." You smiled and said: "Now I know it, too." After looking at the 10 dates and hearing your comments, your administrative assistant wrote down A 's birthday without asking any questions. So what did the assistant write?

Solution.

▷ *Key observation.* Each statement should be interpreted using only the information known at that moment.

First, suppose C could determine the birthday from just the day. Then the day would have to appear only once in the original list. The unique days are

2 and 7,

corresponding to

Dec 2 and Jun 7.

Therefore, if C were told 2 or 7, then C would know the birthday immediately.

You first say:

"I don't know A 's birthday; C doesn't know it either."

Since you are sure that C does not know, your month cannot be June or December. If your month were June, then the birthday could be Jun 7, in which case C would know. Similarly, if your month were December, then the birthday could be Dec 2, in which case C would know.

∴ the month is either March or September.

After hearing this, C says:

"I didn't know A 's birthday, but now I know it."

So C only needs to consider the dates in March or September:

Mar 4, Mar 5, Mar 8, Sep 1, Sep 5.

In this reduced list, the day 5 appears twice. Therefore, if C had been told 5, he still would not know whether the birthday was

Mar 5 or Sep 5.

But C now knows the birthday, so the day cannot be 5. Thus the remaining possibilities are

Mar 4, Mar 8, Sep 1.

Finally, you say:

“Now I know it, too.”

If your month were March, then you would still have two possibilities:

Mar 4 or Mar 8.

So your month cannot be March. Hence the birthday must be

Sep 1.